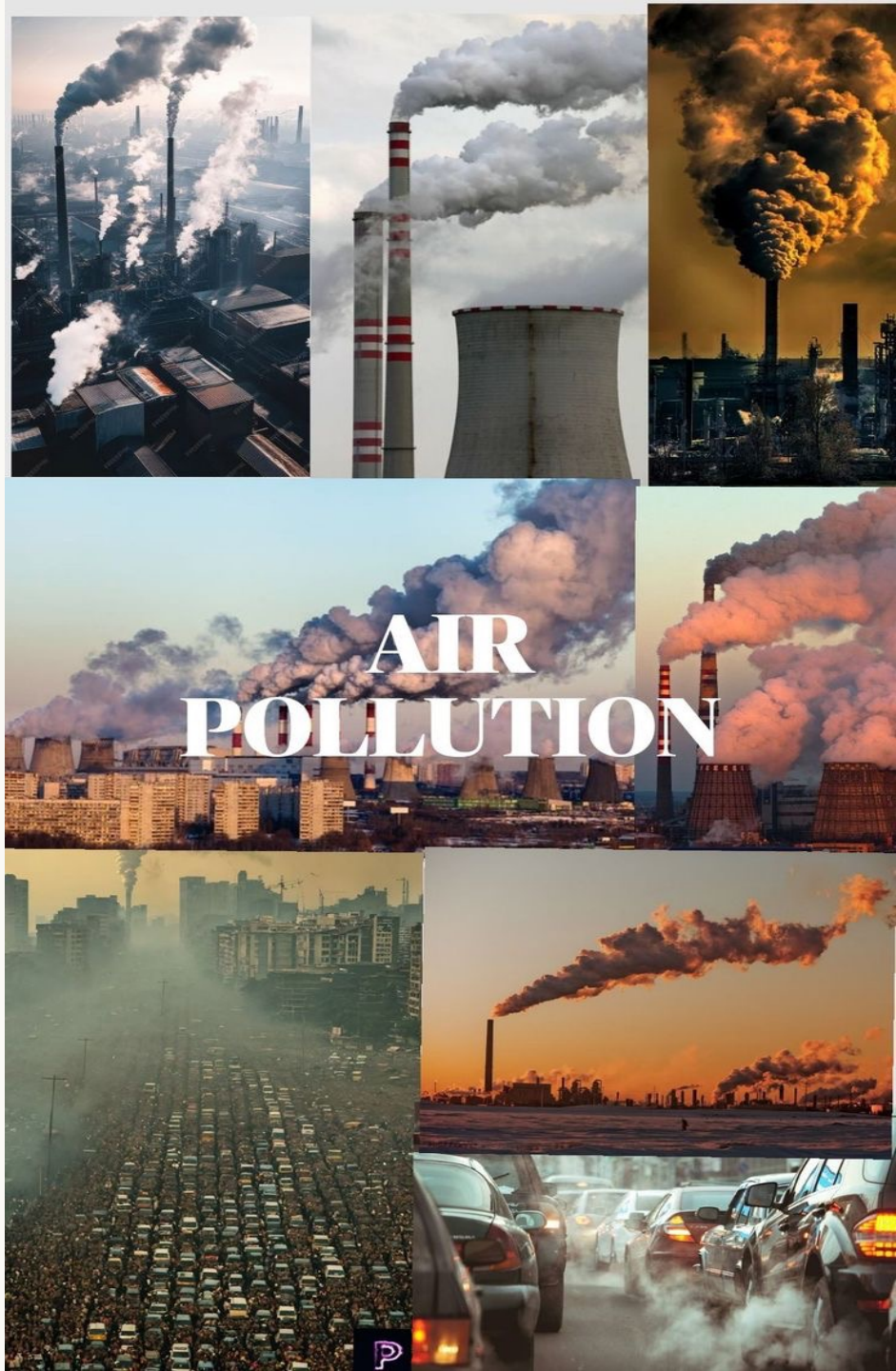
Air Quality Index Prediction and Smart Monitoring System Using Machine Learning

An approach to understanding how air quality affects health risks in urban environments.



Team Members

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BACKGROUND

The Growing Challenge of Urban Air Pollution



Fast urban growth and expanding industries have turned air pollution into a serious public health concern in many Indian cities. Increasing vehicle exhaust, factory emissions, and large-scale construction activities have significantly contributed to worsening air quality levels.

Breathing polluted air for long periods can seriously harm our health, leading to breathing problems, heart diseases, and even a shorter lifespan. Even with these dangers, many people still do not have easy access to simple tools that help them understand daily air quality levels and what they mean for their health.

PROJECT GOALS

Research Objectives and Expected Outcomes

AQI Prediction

Create reliable prediction models using pollution data to estimate upcoming Air Quality Index levels.

Early Warning System

Send early health warnings to at-risk people before air pollution rises to dangerous levels.

Interactive Platform

Design an monitoring system that allows people to check air quality based on different pollution levels.

DATA OVERVIEW

Dataset Characteristics and Sources

Data Collection

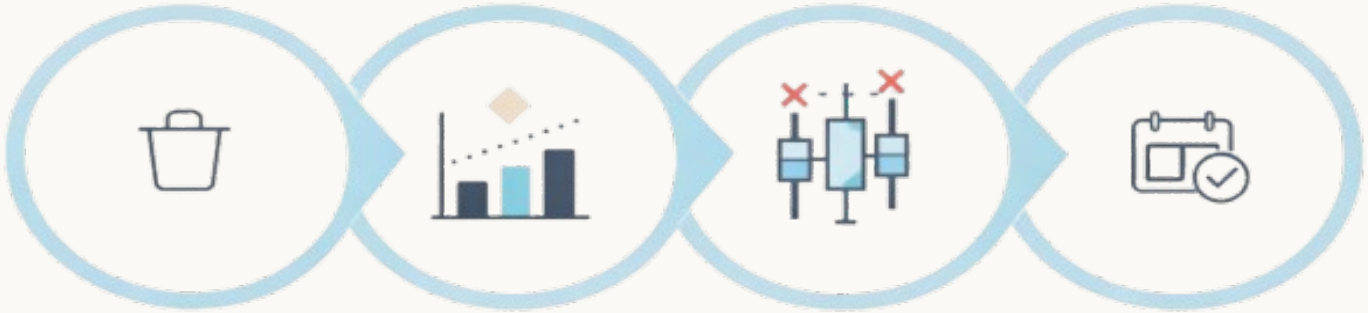
Air quality data for Indian cities (city-day level) from Kaggle: Air Quality Data in India (daily pollution measurements across multiple locations).

Time Period:
Multi-year historical records from year 2015 to 2020

Key Features

- **Particulate Matter:** PM2.5 concentrations (fine particles)
- **Nitrogen Compounds:** NO, NO_2, NO_x levels
- **Carbon Monoxide:** CO measurements
- **Sulphur Dioxide:** SO_2 concentrations
- **Ground-level Ozone:** O_3 readings
- **Target Variable:** Air Quality Index (AQI)

Handling Missing Values and Data Quality



Remove Missing AQI

Median Impute Pollutants

Outlier Removal Using IQR

Date Feature Engineering

AQI Records Removal

Removed rows where Air Quality Index (AQI) was missing, since supervised learning models need correct output labels to learn properly.

Median Imputation Strategy

Used Median values to fill missing pollution data instead of mean as environmental data is often uneven affected by extreme values

Outlier Removal Using IQR

Applied the Interquartile Range(IQR) method to detect and remove extreme pollution values that could negatively impact model performance.

Date Feature Creation

Transformed date data into useful features like month and season to better capture repeating pollution pattern influenced by weather changes and human activities

FEATURES ADDED

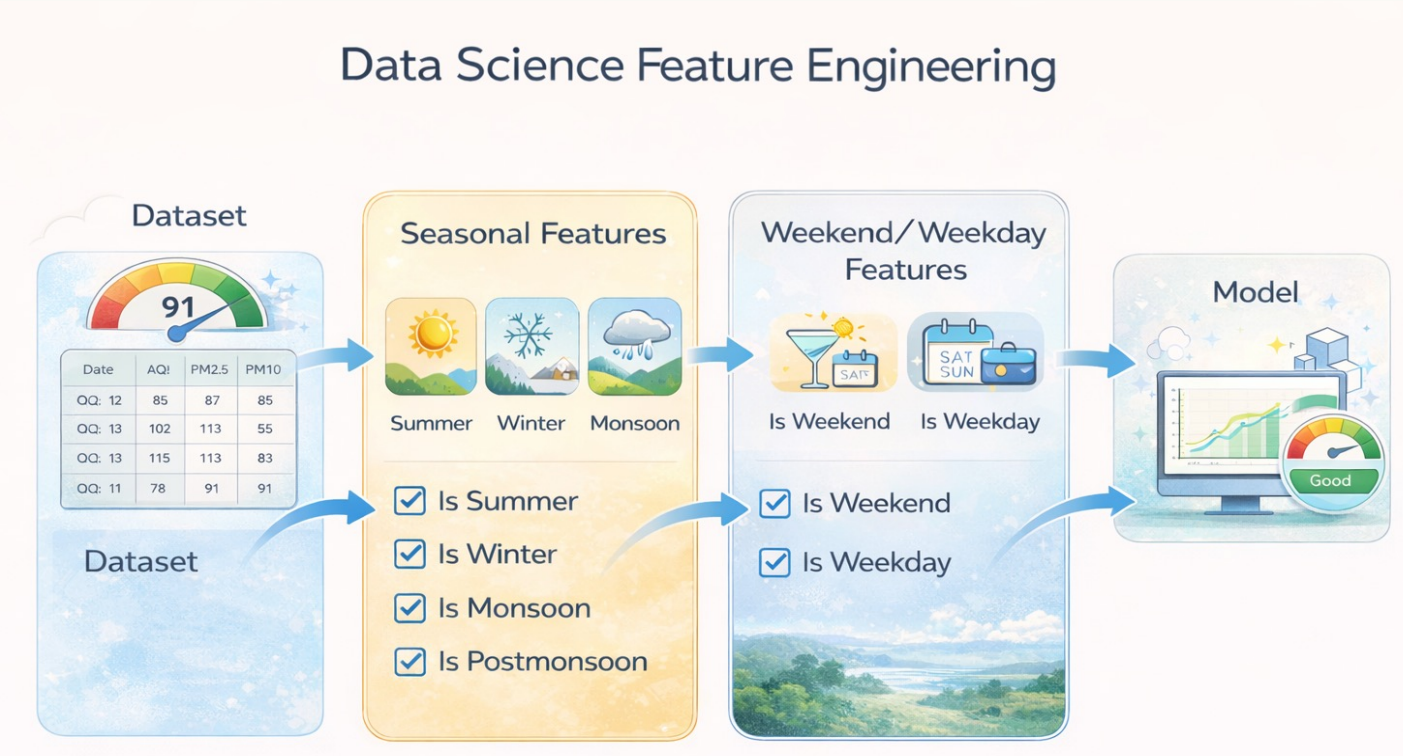
Feature Engineering Approach

Strategic Variable Selection

Feature Engineering was aimed at finding the pollutants that have the strongest statistical relationship with changes in AQI. Through data analysis PM2.5,NO_2 and O_3 were identified as the main factors influencing AQI levels.

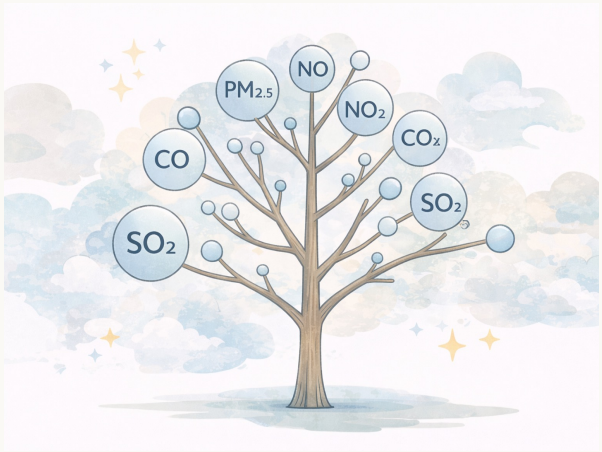
Seasonal Pattern Integration

Added time-based features to reflect seasonal changes in pollution levels. Pollution usually rises during winter because temperature inversions trap pollutants near the ground, while the monsoon season often helps reduce pollution naturally through rainfall.



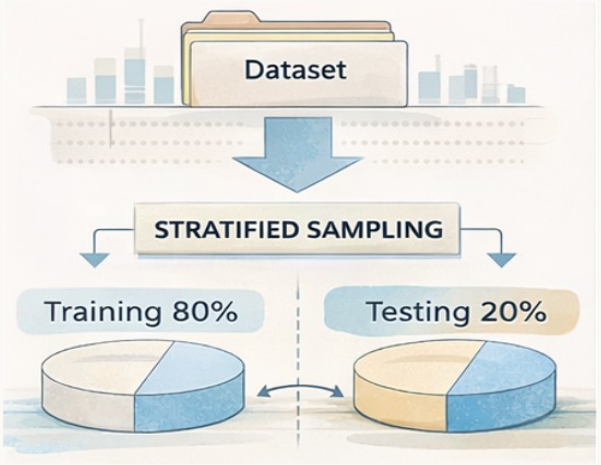
MODEL DEVELOPMENT

Random Forest Regression Architecture



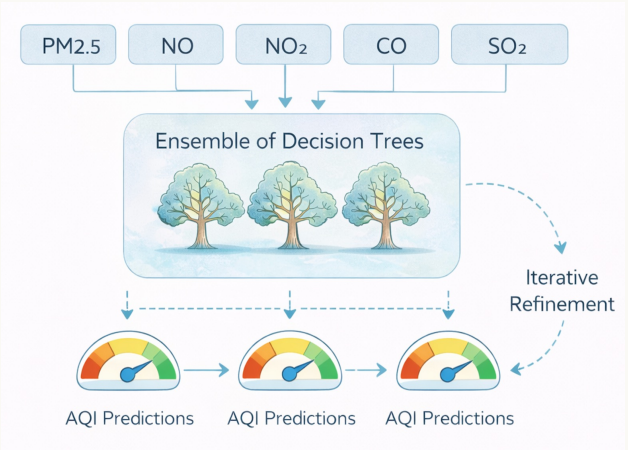
Algorithm Selection

Random Forest was selected because it can understand complex, non-linear relationships between pollutants while reducing the risk of overfitting.



Data Partitioning

We split the data into two parts 80% to train the model and 20% to test how well it learned making sure both parts had a similar mix of data so the results would be reliable.

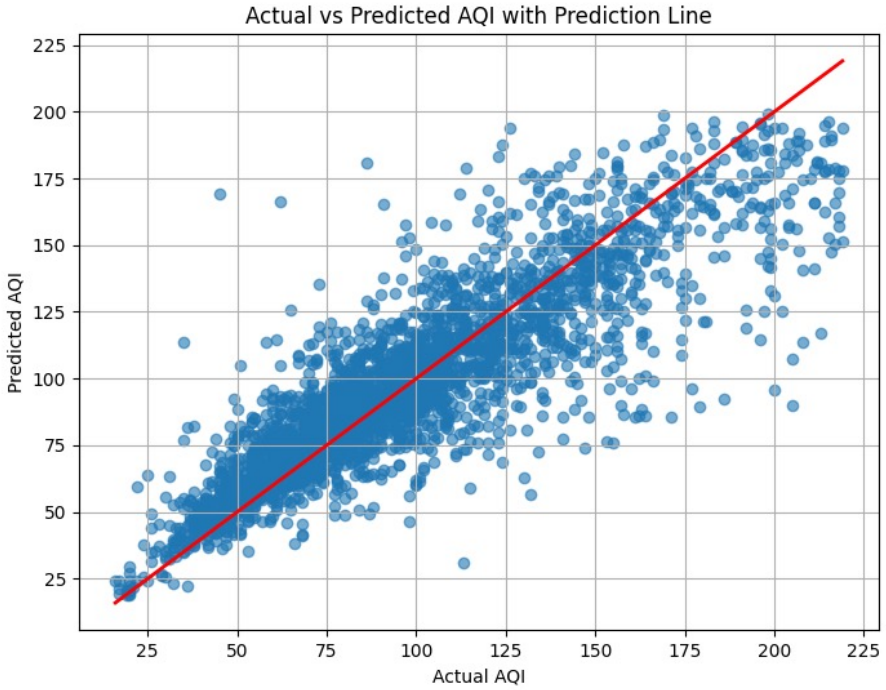


Model Training

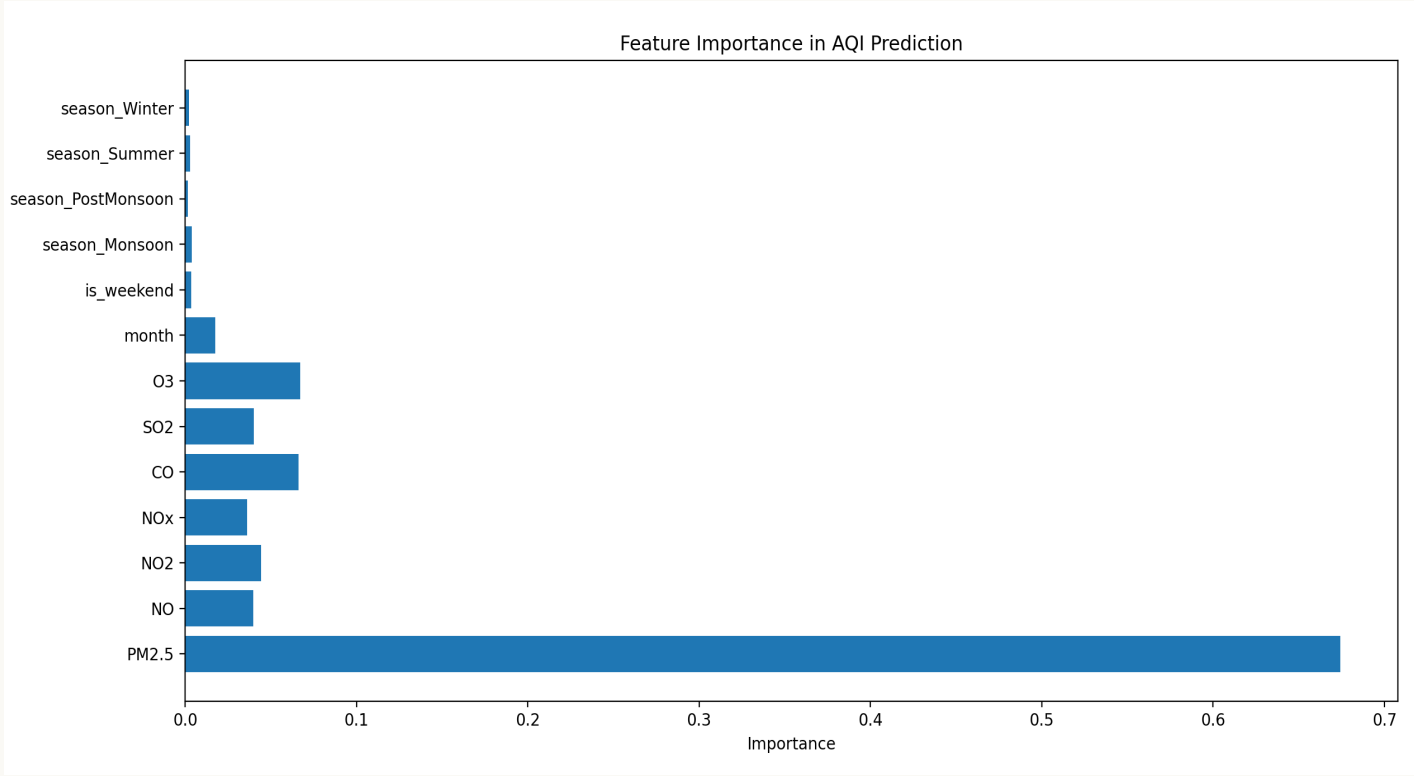
The model gradually learns detailed patterns between pollution levels and the resulting AQI by continuously improving itself with each step.

VISUALIZATION

Model Visualization



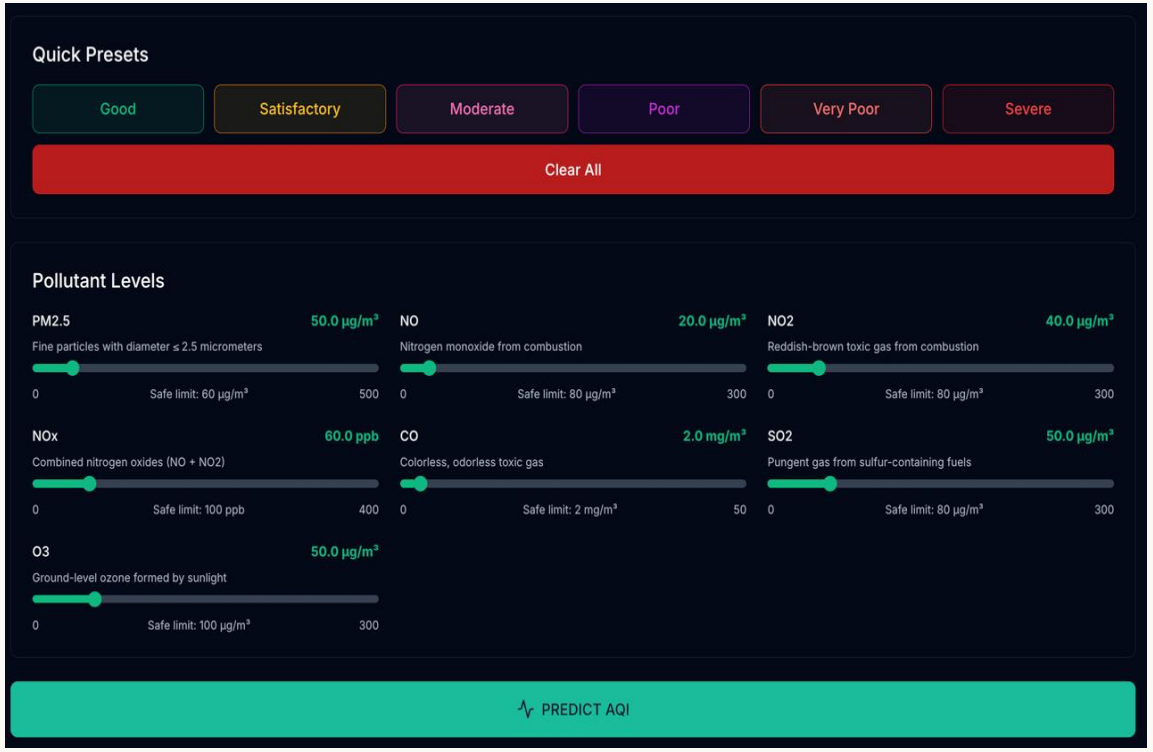
Actual vs Predicted AQI



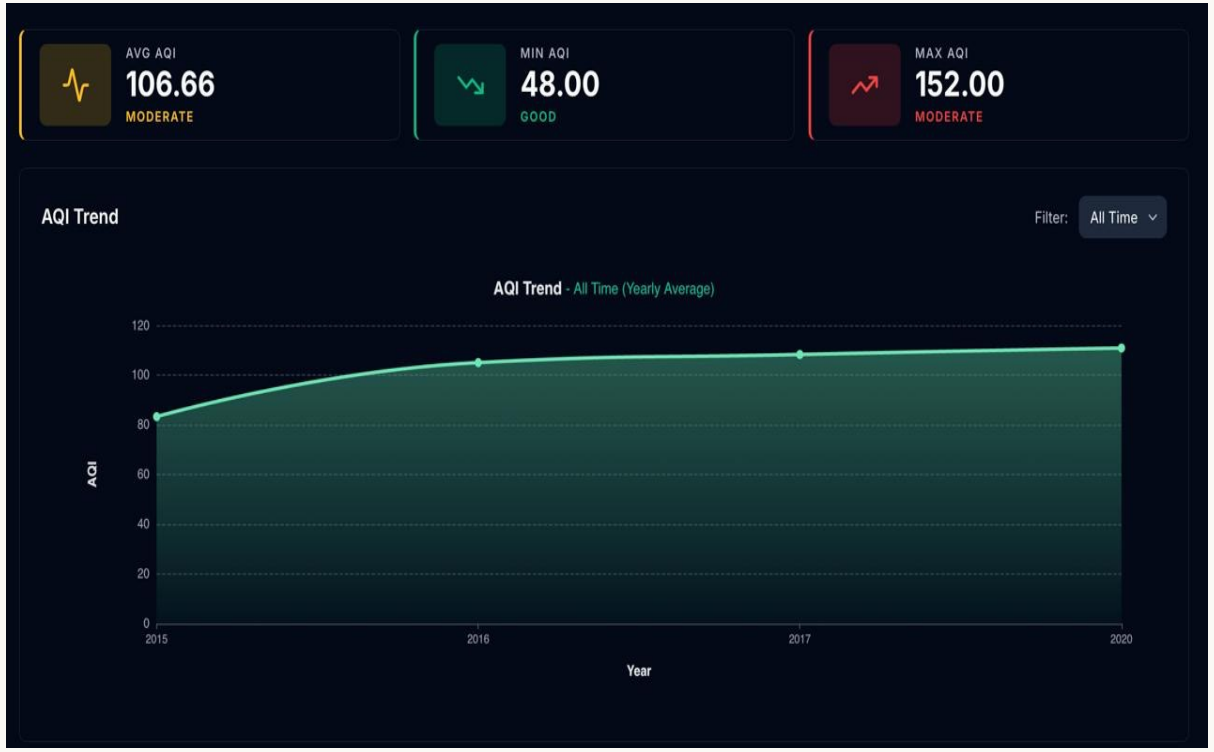
Feature Importance

SYSTEM UI

Building Interactive User Interface



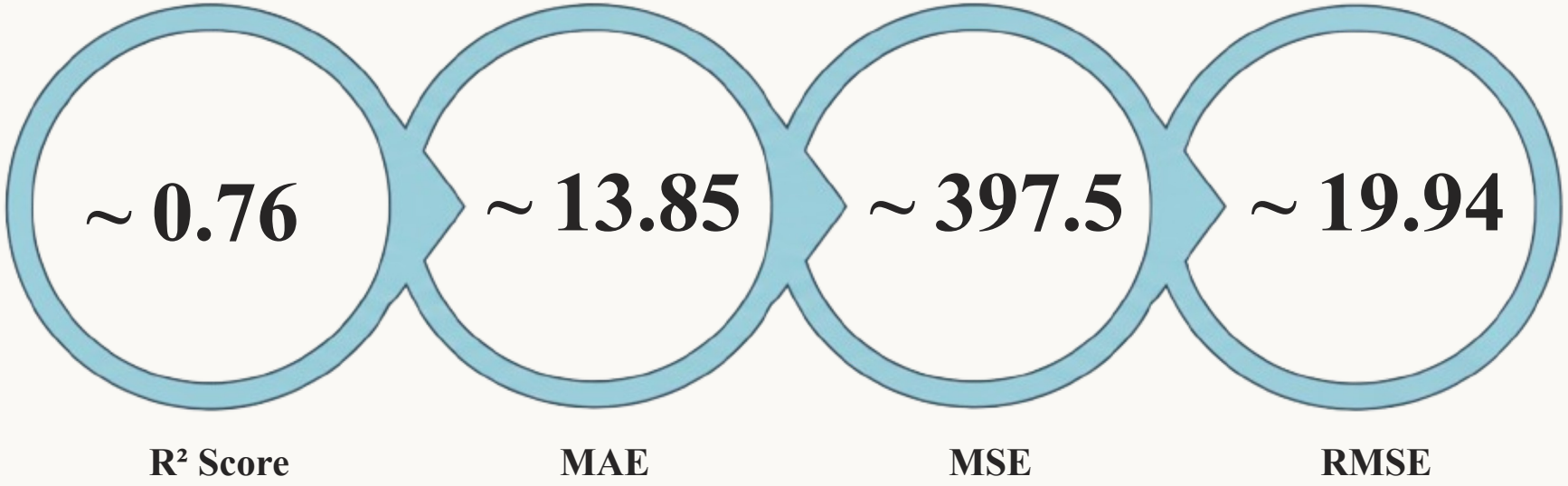
Quick Presets and Pollutants Range Selector for Predicting AQI



Analyzing Dashboard for Monitoring Cities With Time Filter

MODEL PERFORMANCE

Evaluating Model Performance



Our model shows decent performance, with an R^2 score of about 0.76. This means it can explain around 76% of the changes in AQI levels, so it captures most of the important patterns in air quality data.

The MAE value of 13.85 AQI units indicates that, on average, the model's predictions are fairly close to the actual values. The RMSE is around 19.94 AQI units, which suggests that while there are some larger errors, they are not too extreme.

Overall, the results show that the model performs reasonably well and can be useful for analyzing and predicting air quality in real-world situations.

Thanks